ARTIFICIAL INTELLIGENCE | COURSE ASSIGNMENT

Smart Assignment Deadline Tracker Agent

Assignment Prioritization & Deadline Monitoring Agent

Design of an AI Agent: Agent Formulation, Environment Type, Sensors & Actuators, and PEAS Framework Evaluation

SUBMITTED BY	**Saweer U Deen**
SEAT NO	**23122016**
SECTION	**6B**
SUBMITTED TO	**Ma’am Uzma**

Smart Assignment Deadline Tracker Agent

Assignment Prioritization & Deadline Monitoring Agent

1 Problem Statement

Students often struggle to manage multiple assignment deadlines and submit their work on time. Manual deadline tracking can lead to missed submissions, poor time management, and difficulty identifying which assignment requires immediate attention. Students need a simple system that monitors deadlines, tracks completion status, and prioritizes assignments according to their urgency.

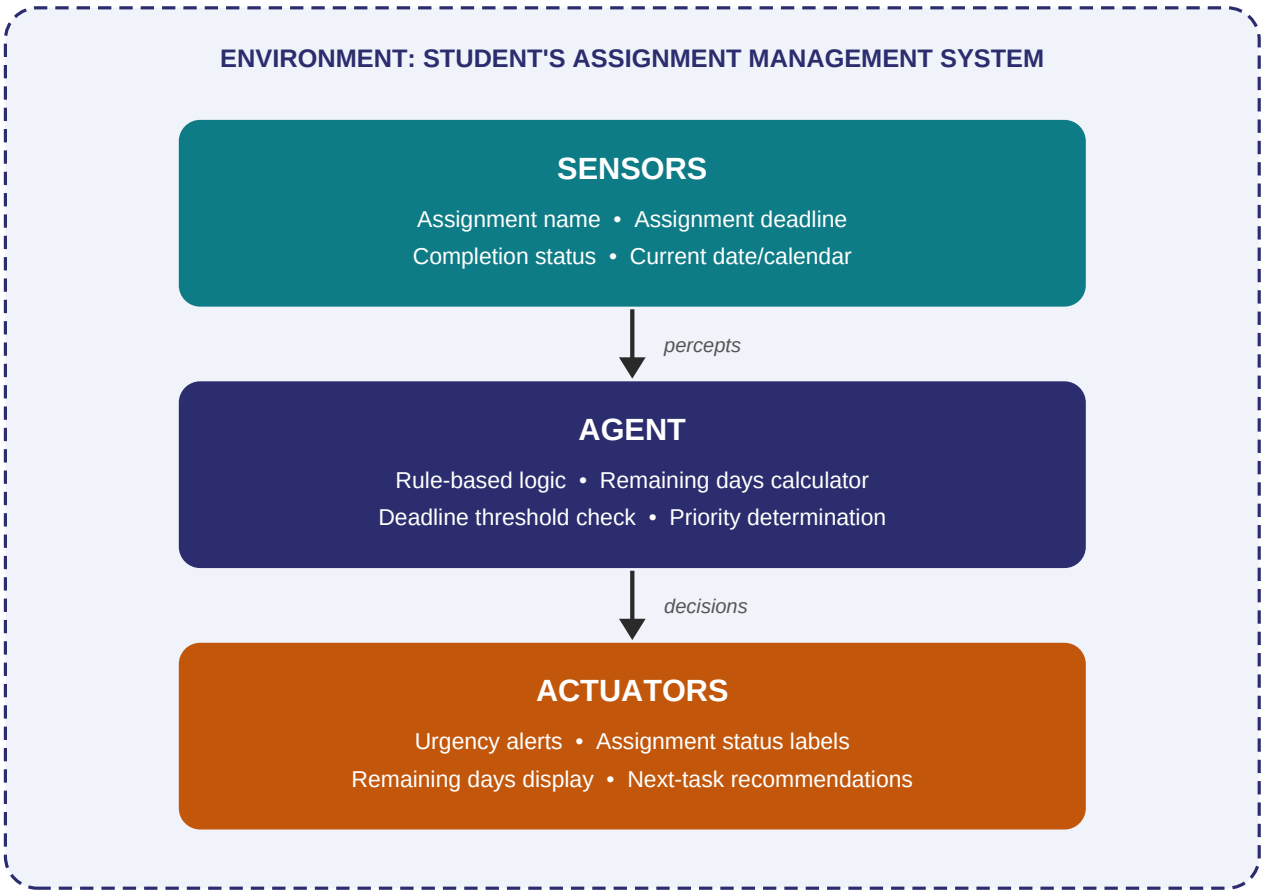
2 Proposed Agent: Smart Assignment Deadline Tracker

The **Smart Assignment Deadline Tracker Agent** takes assignment names, deadlines, completion statuses, and the current date as input. It calculates the remaining days until each deadline, compares the results against predefined rules, and classifies assignments as Urgent, Upcoming, Overdue, or Completed. It also recommends which incomplete assignment should be prioritized.

All decisions are made using simple condition-action rules, making the agent suitable for implementation using basic Python programming.

3 Graphical Representation

The diagram shows how the agent collects assignment information, processes it through rule-based logic, and provides useful results and recommendations to the student.



4 Agent Formulation

AGENT TYPE

The Smart Assignment Deadline Tracker is a **Model-Based Reflex Agent**. It maintains an internal state containing assignment details, deadlines, completion statuses, and remaining days. It applies condition-action rules to classify assignments and recommend priorities based on the available information.

ENVIRONMENT TYPE

Property	Type
Observability	Partially observable
Determinism	Deterministic
Episodic / Sequential	Sequential
Static / Dynamic	Dynamic
Discrete / Continuous	Discrete
Agents	Single-agent

SENSORS

- Assignment names entered by the student
- Assignment deadlines entered by the student
- Assignment completion status
- System date/calendar for calculating remaining days

ACTUATORS

- Urgency alerts and deadline notifications
- Assignment status labels
- Display of remaining days
- Recommendations for the next assignment to complete

5 PEAS Framework

Element	Description
Performance Measure	Accurate deadline tracking, correct urgency classification, reduced missed deadlines, and effective assignment prioritization.
Environment	Student, assignment list, deadlines, current date, and assignment completion statuses.
Actuators	Urgency alerts, assignment status labels, remaining days display, and task-priority recommendations.
Sensors	Assignment names, deadline inputs, completion status, and date/calendar information.